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constexpr containers and adaptors

This paper proposes marking all containers and adaptors `constexpr` (with exception of `std::hive`). This will make compile time programming much easier by making normal C++ and `constexpr` C++ more similar.

This paper doesn't propose changes in `std::hash`.

Revision history

- R2 → R3: updated features test macros
- R1 → R2: Removal of `constexpr` for *`node_handle`*`::mapped()` member function. Removal of `ceilf` mention in the implementation section as the function is constexpr since C++23. Added discussion about hashing and why it's not included. Added informations about *libstdc++* implementability. Extended discussion about change impact on existing code.
- R0 → R1: Added implementation specific problems with `std::less<T*>`.

Introduction and motivation

All necessary `constexpr` functionality required by the containers to be changed is supported by the language. There is no reason for them not to be constexpr, and the standard library should support these in order for them to properly serve as vocabulary types.

The following table shows what this paper proposes, and the status quo:

container / adapter	before	after
`std::array`	✔	✔